Exercise sheet: Week 3 - solutions

1. $\underline{X}$ is a random sample from Exp$(1/\theta)$, with pdf $f_X(x;\theta) = \frac{1}{\theta}\exp(-x/\theta)$.

$$\begin{aligned} U(\theta;\underline{X}) &= \tfrac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{X};\theta) \\ &= -\tfrac{n}{\theta} + \tfrac{\sum X_i}{\theta^2} \\ &= \tfrac{n}{\theta^2}(\overline{X} - \theta) \end{aligned}$$

   From the MVUE Lemma we deduce that

   (i) $\overline{X}$ is the MLE $\widehat{\theta}$ (since this ensures $U = \frac{\partial \ell}{\partial\theta} = 0$)

   (ii) $\overline{X}$ is unbiased (since $\mathbf{E}(U) = 0$)

   (iii) $\overline{X}$ attains the CRLB which is $\frac{1}{n/\theta^2} = \frac{\theta^2}{n}$.

2. (a) $\underline{X}$ is a random sample from Exp(λ) with pdf $f_X(x;\lambda) = \lambda\exp(-\lambda x)$.

$$U(\lambda,\underline{X}) = \frac{\partial}{\partial\lambda}\log f_{\underline{X}}(\underline{X},\theta) = \frac{n}{\lambda} - \sum X_i = n\left(\frac{1}{\lambda} - \overline{X}\right).$$

   Clearly we cannot rearrange $U(\lambda,\underline{X})$ into the form $a(\lambda)(T(\underline{X}) - \lambda)$; hence there is no unbiased estimator of λ that attains the CRLB.

   (b) For the MLE, solve $U = 0 \Rightarrow \frac{1}{\lambda} - \bar{X} = 0 \Rightarrow \widehat{\lambda} = \frac{1}{\overline{X}}$.

   Using hints (proved in tutorial class), since $\overline{X} \sim \Gamma(n, n\lambda)$, then $\mathbf{E}(\overline{X}^{-1}) = \frac{\Gamma(n-1)}{\Gamma(n)}n\lambda = \frac{n}{n-1}\lambda$ (we require $n > 1$).

   Hence $\mathbf{E}(\widehat{\lambda}) \neq \lambda$, so that $\widehat{\lambda}$ is biased. Clearly $\widetilde{\lambda} = \frac{n-1}{n}\hat{\lambda}$ is unbiased.

   (c) $\mathbf{Var}(\widehat{\lambda}) = \mathbf{E}(\widehat{\lambda}^2) - (\mathbf{E}(\widehat{\lambda}))^2$.

   Now $\mathbf{E}(\widehat{\lambda}^2) = \mathbf{E}(\overline{X}^{-2}) = \frac{\Gamma(n-2)}{\Gamma(n)}(n\lambda)^2 = \frac{n^2}{(n-1)(n-2)}\lambda^2$ (for $n > 2$)
   $\Rightarrow \mathbf{Var}(\widehat{\lambda}) = \frac{n^2}{(n-1)(n-2)}\lambda^2 - \frac{n^2}{(n-1)^2}\lambda^2 = \frac{n^2}{(n-1)^2(n-2)}\lambda^2$.

   Also $\mathbf{Var}(c\widehat{\lambda}) = c^2\frac{n^2}{(n-1)^2(n-2)}\lambda^2$.

$$\text{CRLB} = \frac{1}{\mathbf{E}(U^2)} = \frac{1}{n^2\mathbf{E}\left(\left(\overline{X} - \frac{1}{\lambda}\right)^2\right)} = \frac{1}{n^2\mathbf{Var}(\overline{X})},$$

   since $\mathbf{E}(\overline{X}) = \frac{1}{\lambda}$.

   Since $\mathbf{Var}(\overline{X}) = \frac{1}{n} \times \frac{1}{\lambda^2}$ $\left(= \frac{\sigma^2}{n}\right)$, it follows that CRLB = $\frac{\lambda^2}{n}$.

   We can see that $\mathbf{Var}(\widehat{\lambda}) > \mathbf{Var}(\widetilde{\lambda}) >$ CRLB $= \frac{\lambda^2}{n}$. As $n \to \infty$, $\frac{\mathbf{Var}(\widehat{\lambda})}{\text{CRLB}}$ and $\frac{\mathbf{Var}(\widetilde{\lambda})}{\text{CRLB}}$ both $\to 1$.

   Moreover, as $n \to \infty$, $\mathbf{E}(\widehat{\lambda}) = \frac{n}{n-1}\lambda \to \lambda$, so that $\widehat{\lambda}$ is asymptotically unbiased.

   Hence, both $\widehat{\lambda}$ and $\widetilde{\lambda}$ are asymptotically MVUEs (Minimum Variance Unbiased Estimators).

3. $\underline{X}$ is a random sample from a distribution with pdf $f_X(x;\sigma^2) = \frac{x}{\sigma^2}\exp\left(-\frac{x^2}{2\sigma^2}\right)$, $(x>0)$

(a) The case $\theta = \sigma^2$:

$$f_{\underline{X}}(\underline{X};\theta) = \prod_{i=1}^{n}\frac{X_i}{\theta}\exp\left(-\frac{X_i^2}{2\theta}\right) = \left(\prod_{i=1}^{n}X_i\right)\theta^{-n}\exp\left(-\frac{\sum X_i^2}{2\theta}\right).$$

NB: This implies that $\sum X_i^2$ is sufficient for θ.

$$U(\theta;\underline{X}) = \frac{\partial}{\partial\theta}\ell(\theta) = -\frac{n}{\theta} + \frac{\sum X_i^2}{2\theta^2} = \frac{n}{\theta^2}\left(\frac{\sum X_i^2}{2n} - \theta\right). \tag{1}$$

From the MVUE Lemma we deduce that i) $\widehat{\theta} = \frac{\sum X_i^2}{2n}$ is the MLE of θ; ii) $\widehat{\theta}$ is unbiased; iii) $\widehat{\theta}$ attains the CRLB, which is $\frac{\theta^2}{n}$.

(b) The case $\theta = \sigma$:

Replacing θ by θ^2 in the above, we obtain

$$\ell(\theta) = \text{constant} - n\log(\theta^2) - \frac{\sum X_i^2}{2\theta^2}$$

$\Rightarrow U(\theta;\underline{X}) = -\frac{2n}{\theta} + \frac{\sum X_i^2}{\theta^3} = \frac{2n}{\theta^2}\left(\frac{\sum X_i^2}{2n\theta} - \theta\right)$.

This cannot be written as $a(\theta)(T(\mathbf{X}) - \theta)$ $\Rightarrow$ CRLB cannot be attained.

To find the CRLB we require $\mathbf{E}(U^2)$ or (easier in this case) $E\left[\frac{\partial^2\ell(\theta)}{\partial\theta^2}\right]$:

$$\frac{\partial^2\ell(\theta)}{\partial\theta^2} = \frac{2n}{\theta^2} - \frac{3\sum X_i^2}{\theta^4} \qquad \Rightarrow \qquad \mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\ell(\theta;\mathbf{X})\right) = \frac{2n}{\theta^2} - \frac{3\mathbf{E}(\sum X_i^2)}{\theta^4}.$$

From (1) above, and noting that in (b) we have replaced θ by θ^2, it follows that $\mathbf{E}(X_i^2) = 2\theta^2$, so that

$$\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{X})\right) = \frac{2n}{\theta^2} - \frac{3n2\theta^2}{\theta^4} = -\frac{4n}{\theta^2}.$$

It follows that the CRLB = $-\frac{1}{(-4n/\theta^2)} = \frac{\theta^2}{4n}$.